

$$C_{ik} = P(x_i | z_i = k, \theta^{(t)})$$

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^N \sum_{k=1}^K \left[\ln(\pi_k) - \frac{d}{2} \ln(2\pi) - \frac{d}{2} \ln(\sigma_k^2) - \frac{1}{\sigma_k^2} \|x_i - \mu_k\|^2 \right] C_{ik}$$

~~$$\frac{\partial Q}{\partial \pi_k} = 0$$~~

because it will NOT
meet the constraints

on π_k .

$$\sum_{k=1}^K \pi_k = 1$$

Assume $K=2$:



region that maximizes $\{\mu_k, \sum_k\}$

$$\sum_{k=1}^2 \pi_k = 1 \Leftrightarrow 1 - \sum_{k=1}^K \pi_k = 0$$

by convention, we are max'ing "Add".

$$Q_{\pi}(\theta, \theta^t, \lambda) = Q(\theta, \theta^t) + \lambda \left(1 - \sum_{k=1}^K \pi_k \right)$$

Lagrangian fct.

$$\frac{\partial Q}{\partial \pi_k} = 0 \Leftrightarrow$$

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N C_{ik}$$